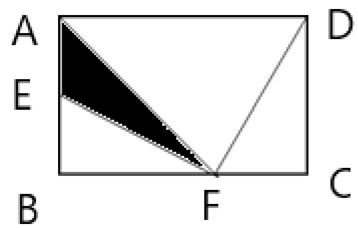
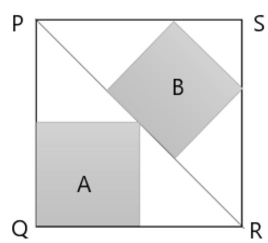
JMO – 2024 (19^{th} JAN-2025)

Answer all questions. All questions carry equal marks.
Full Marks 100

1. Evaluate: $\frac{1}{28}+\frac{1}{36}+\frac{1}{45}+\cdots+\frac{1}{4950}$
2. $a+13b$ is divisible by 11 and $a+11b$ is divisible by 13. Then find the minimum value of $a+b$.
3. How many positive integral multiple of three les than 1000 are there the sum of whose digits is divisible by 7 ?
4. ABCD is a rectangle in which AD = 21 cm, BF = CD, EF = DF = 15 cm, and $m\angle EFD = 90^0$. Find the area of shaded portion.

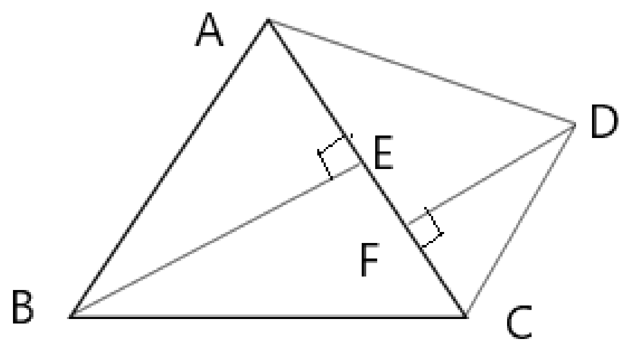


5. Find the sum of 11+192+1993+19994+…+1999999999 without direct addition.
6. Two squares A and B are embedded in a square PQRS. If the side of the square PQRS is 12 cm. Find the area of the shaded portion.

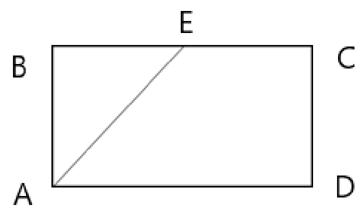


7. If $x+y+z=1$, where x, y, z are any distinct positive real numbers, then prove that $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}>3$.
8. Find the smallest number which when divided by 71 gives remainder 52 and when the said number is divided by 73 gives the remainder 24.
9. Solve for x and y: $3x^2-8xy+9y^2-4x+6y+13=0$.
10. If $(a^2+b^2)^3=(a^3+b^3)^2$, then find the value of $\left(\frac{a}{b}+\frac{b}{a}\right)$.
11. If a, b, c and d are natural numbers and $a+b+c+d=27$, then find the maximum value of $ab+bc+cd+da$.

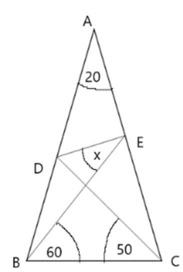
12. In the figure ABCD is a quadrilateral. Diagonal AC = 77 cm, AF = CE = 45 cm, $\overline{BE} \perp \overline{AC}$ and $\overline{DF} \perp \overline{AC}$. If BE = 60 cm and DF = 24 cm. Find the length of diagonal $\overline{BD}$.



13. What is the 2025th term of the sequence of numbers with terms greater than 1 and divisible by 3 or 7 ?
14. ABCD is a rectangle in which $AB + BC + CD = 20\,cm$ and AE = 9 cm. Where E is the mid point of the side BC. Find the area of the rectangle.



15. A bag contains 11 red, 7 blue, 9 green and 3 black marbles. How many marbles must be taken out from the bag at a time so as to get at least 5 marbles of same colour in collected sample?
16. At what time between 7 AM to 8 AM the minute hand is 9 minute ahead of the hour hand?
17. In triangle ABC, AB = AC, $m\angle A = 20^0$, D and E are respectively two such points on AB and AC that $m\angle EBC = 60^0$ and $m\angle DCB = 50^0$. Then find $m\angle DEB$.



18. The grandson asked his grandfather how old he is? The grandfather replied, "I am 72 years old. But I do not compute all Sundays passed during my life time." Find the actual age of his grandfather.

19. If p, q, r, and s are real numbers, then find the least value of :

$$(p^2 + q^2 + r^2 + s^2) - (p + q + r + s)$$

20. A car travels from A to B with its initial speed within a stipulated time. If the speed of the car is 6 km/hr more, then it can reach B 4 hours before the stipulated time. If the speed is 6 km/hr less than the initial speed, then it will take 6 hours more to reach the destination. Find the distance from A to

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